How Preprocessing Choices Impact Sentiment Classification: A Case Study of Formula 1 Social Media

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***Abstract*—Formula One (F1) is the highest class of global motor racing for open-wheel single-seater formula cars fun by the Formula One Group and sanctioned by the Federation Internationale de L'Automobile (FiA). Sentiment Analysis is a research area in Natural Language Processing (NLP) that focuses on extracting semantic and emotional inferences from natural language, paving the way for analyzes that deal with a large volume of textual data. The importance of data in strategic decision-making cannot be overstated and the recognition of social networks as sources of public opinion has significantly aided this study, which aims to explore the interaction between human emotions and F1 events. This study used VADER, a popular model used in sentiment analysis implemented on a public dataset of 650,000+ tweets collected during the 2021 and 2022 F1 seasons.**

Key finding: Preprocessing impact is asymmetric. Standard preprocessing (removing URLs/emojis) caused a 0.0333-point average sentiment divergence compared to minimal preprocessing, representing a meaningful difference in classification. However, entity masking had negligible impact (0.0007-point divergence), suggesting VADER sentiment analysis is robust to entity substitution but sensitive to affective signals (emojis). Team-specific analysis revealed significant differences in public perception: Ferrari maintained the most positive sentiment (+0.1941 mean), Red Bull occupied middle ground (+0.1598), and Mercedes showed greater polarization (+0.1363 with 26.4% negative mentions). Temporal Analysis identified three major sentiment pivot points corresponding to regulatory changes and on-track incidents. Results demonstrate that preprocessing choices significantly impact sentiment conclusions, with implications for NLP practitioners, sports analytics, and social listening applications. The findings revealed significant engagement patterns, with notable spikes in emotional reactions coinciding with critical moments of the seasons.

I. Introduction

Not long ago, F1 was a niche European sport followed mainly by purists and automotive enthusiasts. Today, it is a cultural phenomenon with record-breaking viewership numbers, celebrity appearances, memes & clips, race highlights and so on. Factors that have transformed the sport into one of the most powerful entertainment trends. With F1 continuing to grow in today's environment, the number of online conversations centered around F1 have been on the rise as well. This project aims to analyze the way different preprocessing approaches impacted tweets from 2021-2022. Team loyalty in F1 is fiercely tribal with sentiments shifting rapidly after each race weekend, I aim to investigate how preprocessing choices affect my results.

In the past decade, Machine Learning (ML) has continuously reshaped our interactions with data [1]. NLP has been a pivotal advancement in how we process and interpret human language, marking a revolutionary step in the realm of computational linguistics and Artificial Intelligence (AI) [2]. Social media has become a rich real-time source of data where users voice opinions and insights on various topics. Analyzing these sentiments provide a valuable understanding of the public perception, which is applicable to various areas, including sports, politics, entertainment, science and marketing [3].

This study examines Twitter engagement patterns during the 2021–2022 Formula 1 seasons to decode the motivations behind fan sentiment. The findings provide dual benefit: offering strategic intelligence on public perception and establishing a framework for how team performance and regulatory decisions influence collective emotions. From a detailed analysis of expressed sentiments, valuable data carefully be obtained that can be leveraged for developing more effective marketing and communication strategies in the sport. Ultimately, this study addresses two research questions:

- How does preprocessing strategy impact sentiment classification in F1 social media?
- How has public perception of F1 teams evolved, and can NLP identify pivot points where sentiment shifted?

II. Related Works

Sentiment analysis on Twitter has been a prominent research area, with key studies employing various methodologies and machine learning techniques. Go et al. [4] and Pak et al. [5] were early contributors, focusing on automatic sentiment classification using distant supervision and linguistic analysis of tweets, respectively. Both studies utilized Naive Bayes classifiers and emphasized the importance of preprocessing and feature extraction techniques such as n-grams.

In dataset analysis, Saif et al. [6] and Koto and Adriani [7] provided insights into the performance correlations between dataset characteristics and sentiment classification. They explored various textual attributes, including opinion and sentiment lexicons such as AFINN and Senti-Strength, and found that some widely known lexicons like SentiWordNet were less effective for Twitter sentiment analysis.

Using the same dataset I used for this study, Mileski et al. [8] using advanced Machine Learning (ML) and Deep Learning techniques (DL) like Logistic Regression and Long Short Term Memory (LSTM) to classify the sentiments expressed by fans during the 2021-2022 season. Their studies showed how

the accuracy between simple ML and advanced DL techniques differ in sentiment analysis and classification.

III. Methodology

A. Dataset

This study used a textual dataset titled, "Formula 1 (F1) trending tweets", which is currently available on Kaggle. The dataset covers tweets posted with the hashtag "F1" in the period from July 24, 2021, to August 20, 2022. It has over 600,000 tweets and there was no specific methodology used to collect the data. Figure 1 displays the features available in the F1 dataset. Mainly the text and date features were used for this study. The text column is essential for applying sentiment analysis models, while the date column allows one to perform an exploratory analysis of sentiments about F1 events.

Feature	Description
user_name	Username as defined by the user
user_location	User defined location for profile
user_description	User-defined description for your account
user_created	Date and time the user account was created
user_followers	Number of user followers
user_friends	Number of user friends
user_favorites	Number of user favorites
user_verified	Whether the account is verified or not
date	UTC date and time of posting the tweet
text	Text from tweet
hashtags	Additional hashtags in tweet besides #f1
source	Tool used to post tweet
is_retweet	Indicates whether the tweet was reposted by the authenticated user

Fig. 1: F1 Dataset Features

B. Preprocessing

Three different preprocessing approaches were used in this study: Minimal, Standard and F1-Aware. Each strategy represents a defensible choice commonly used in NLP practice, allowing systematic comparison of their impact on sentiment classification.

- **Minimal Processing:** Minimal preprocessing preserves the original tweet as much as possible, serving as a baseline for comparison. This approach only removes null values and converts the text to string format. All other elements are retained: URLs, @mentions, emojis, hashtags, punctuation, and original capitalization. This strategy tests sentiment analysis on raw social media text, which reflects how humans experience the tweets.
- **Standard Preprocessing:** Standard preprocessing represents common NLP practice and involves removing URLs and @mentions with Regex, as well as emojis/special characters. Text is converted to lowercase for normalization. Importantly, hashtags are preserved because they carry semantic meaning in F1 context, #AbuDhabiGP identifies a specific race, #SilverArrows identifies Mercedes, and #DriveToSurvive indicates Netflix-related discussion. These are content signals, not noise. This strategy tests whether removing URLs, mentions, and emojis significantly impacts sentiment conclusions.
- **F1-Aware Preprocessing:** F1-Aware preprocessing applies all standard preprocessing steps, then adds entity masking: driver names (Hamilton, Verstappen, Leclerc, Sainz, Norris, etc.) are replaced with the generic token 'DRIVER', and team names (Mercedes, Ferrari, Red Bull, etc.) are replaced with 'TEAM'. Entity masking tests whether sentiment is entity-dependent (specific to a driver or team) or context-dependent (driven by words like 'amazing', 'terrible', dominated'). If entity masking causes minimal sentiment divergence, it suggests VADER operates primarily on context words rather than proper nouns.

After these cleaning steps, the texts were subjected to tokenisation at the word and sub-word level. Words considered *"stopwords"* were removed to reduce noise and focus on meaningful words for analysis [9].

Preprocessing Example

Original: Breaking news FIA has disqualified @OconEsteban for running down the pit lane too fast @LewisHamilton takes the win #F1 #HungaryGP #BREAKINGNEWS #FIA

- **Strategy A (Minimal)**: Breaking news FIA has disqualified @OconEsteban for running down the pit lane too fast @LewisHamilton takes the win #F1 #HungaryGP #BREAKINGNEWS #FIA
- **Strategy B (Standard)**: breaking news fia has disqualified for running down the pit lane too fast takes the win #f1 #hungarygp #breakingnews #fia
- **Strategy C (F1-Aware)**: breaking news fia has disqualified ⟨user⟩for running down the pit lane too fast ⟨user⟩takes the win #f1 #hungarygp #breakingnews #fia

Note: In some tweets, driver names appear as @mentions (e.g., @LewisHamilton). These are removed during Standard preprocessing as part of @mention removal, and therefore are not available for entity masking in Strategy C. Entity masking in Strategy C applies only to driver/team names that appear as regular text (e.g. "Hamilton", "Mercedes") rather than @mentions.

C. Sentiment Analysis (VADER)

Sentiment analysis was performed using VADER(Valence Aware Dictionary and sEntiment Reasoner), a lexicon-based sentiment analyzer specifically trained on social media text. VADER computes sentiment by tokenising input text, matching tokens against a lexicon of 7,500 words with pre-assigned sentiment polarities, and adjusting scores for intensifiers, negations, punctuation, and capitalization. VADER outputs four scores: negative proportion, positive proportion, neutral proportion, and compound score (ranging from -1 to +1, where -1 is most negative and +1 is most positive).

I used the compound score as our primary sentiment metric. Classification rules were applied consistently across all preprocessing strategies:

- Positive sentiment: compound <0.05
- Negative sentiment: compound ≤ -0.05
- Neutral sentiment: -0.05 compound ≤ 0.05

VADER was selected because it requires no fine-tuning, works out-of-the-box on Twitter data, and provides interpretable decisions based on lexicon matching rather than black-box deep learning.

D. Tokenisation and Stopword Removal

After sentiment analysis, tweets were tokenised at the word level using the `nltk`'s `word_tokenize` function. Stopwords (common words like 'the', 'a', 'is') from the English stopwords corpus were removed to reduce noise and focus analysis on semantically meaningful terms. This tokenisation was applied for downstream analysis of token frequency and language patterns by sentiment category.

E. Team-Specific Sentiment Extraction

To analyze team-specific sentiment, tweets mentioning specific teams were identified using keyword matching. Keywords for each team were:

- **Mercedes:** ['mercedes', 'merc', 'silver arrows', 'hamilton', 'lewis', 'lh']
- **Red Bull:** ['red bull', 'redbull', 'rb', 'max verstappen', 'max', 'checo', 'sergio', 'perez']

Tweets matching any keyword in a team's list were classified as mentions of that team. Team-specific sentiment was calculated only from these filtered tweets.

F. Temporal Analysis

Tweets were aggregated by month to create temporal sentiment trends. Monthly mean sentiment was computed for each team. Rolling window smoothing (window = 2 months, center-aligned) was applied to reduce noise and identify underlying trends.

Pivot points (significant sentiment shifts) were identified where month-to-month sentiment change exceeded ±0.1 points in absolute value. This threshold was chosen empirically as a meaningful signal above daily fluctuations. Pivot points were cross-referenced with real-world F1 events (regulatory changes, championship incidents, driver transfers) to assess whether NLP could detect public reaction to significant events.

G. Statistical Analysis

Descriptive statistics (mean, median, standard deviation) were computed for sentiment distributions. Preprocessing impact was quantified as absolute divergence:

$$mean\left(|sentiment_{strategyA} - sentiment_{strategyB}|\right).$$

Polarity flip rate (percentage of tweets where classification changed between strategies) was also calculated.

All analyses were performed using Python 3 (via Google Colab) with `pandas`, `matplotlib`, `nltk`, and VADER sentiment.

IV. RESULTS

This section presents findings from three main analyses: (1) preprocessing impact on sentiment classification, (2) team-specific sentiment rankings and trends, and (3) temporal sentiment patterns and pivot point detection.

A. Preprocessing Impact Analysis

1) Sentiment Score Distributions: The three preprocessing strategies produced measurably different sentiment distributions across the 600,000+ tweets. Descriptive statistics are presented in Table I.

TABLE I: Sentiment Score Comparison

	Minimal	Standard	F1-Aware
count	632388.000000	632388.000000	632388.000000
mean	0.148214	0.142883	0.143333
std	0.436712	0.430669	0.431026
min	-0.997600	-0.997500	-0.997500
25%	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000
75%	0.510600	0.493900	0.493900
max	0.998800	0.997500	0.997500

All three strategies show similar overall distributions (medians all at 0, similar standard deviations). However, subtle differences in mean sentiment and quartiles suggest preprocessing choices do affect the distribution shape.

2) Divergence Analysis: To quantify how much each preprocessing strategy changed results, The absolute divergence was computed:

$$\text{mean}\ |sentiment_{strategyA} - sentiment_{strategyB}|.$$

Results are in Table II below.

TABLE II: Mean Absolute Divergence Comparison

Comparison	Mean Absolute Divergence	Interpretation
Minimal vs Standard	0.0333	Moderate divergence
Minimal vs F1-Aware	0.0337	Moderate divergence
Standard vs F1-Aware	0.0007	Negligible divergence

Removing URLs, @mentions, and emojis (Minimal → Standard) caused a 0.0333-point average sentiment change, representing a 2.3% shift in the sentiment scale. However, entity masking (Standard → F1-Aware) caused only 0.0007-point divergence, suggesting that replacing driver/team names with generic tokens has minimal impact on sentiment classification.

The implication is that emojis matter more for VADER sentiment analysis than entity identity.

B. Team Sentiment Analysis

1) Overall Team Sentiment Rankings: Team-specific sentiment was extracted from tweets mentioning Mercedes, Red Bull, or Ferrari (using keyword matching). Results are presented in Table III

TABLE III: Team Sentiment Summary

Team	Tweet Count	Mean Sentiment	Median	Std Dev	% Positive
Ferrari	18,973	+0.1941	+0.0258	0.4209	49.9%
Red Bull	56,623	+0.1598	0.0000	0.4591	47.8%
Mercedes	46,101	+0.1363	0.0000	0.4552	47.3%

- Ferrari maintained most positive sentiment (+0.1941), with nearly 50% positive mentions. This may reflect strong fan loyalty and/or less recent championship pressure.
- Mercedes showed greatest polarization (+0.1363 mean but 26.4% negative), the highest negative mention ratio.

This suggests fan frustration, possibly related to [contextual explanation: controversial decisions at the end of the 2021 F1 season and a drop in performance in the 2022 season].

- Red Bull occupied middle position (+0.1598), with balanced positive (47.8%) and negative (23.3%) mentions, reflecting competitive team status.

2) Sentiment Distribution by Team: Figure 2 displays the sentiment distribution for each team, showing the proportion of positive, negative, and neutral mentions.

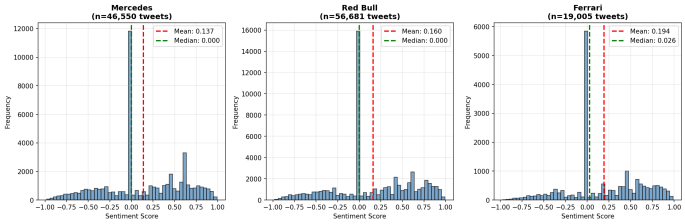

Fig. 2: Team Sentiment Distributions

C. Temporal Sentiment Analysis

1) Overall Sentiment Trends (2021-2022): Tweets were aggregated by month to examine how overall F1 sentiment evolved over the study period. Figure 3 displays monthly mean sentiment with ±1 standard deviation shading.

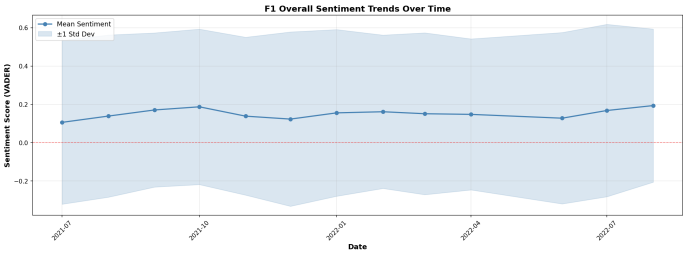

Fig. 3: Overall Sentiment Trends Over Time

The overall F1 sentiment remained positive throughout 2021-2022, with a modest upward trend (+0.04 increase over 13 months). This suggests growing fan engagement and satisfaction despite championship controversies and regulation changes across that period.

2) Team-Specific Sentiment Trends: Figure 4 shows monthly mean sentiment for each team over time, with team-specific trends highlighted.

- Ferrari maintained consistently positive sentiment (+0.15 to +0.25), with little volatility.
- Mercedes showed greater fluctuation (+0.08 to +0.18) with notable dips at the end of 2021, likely corresponding to controversy at the Abu Dhabi GP at the time which was held in December.
- Red Bull remained relatively stable (+0.12 to +0.18) throughout the period.

D. Pivot Point Detection

Figure 5 highlights dramatic shift in sentiments (>0.1 points month-to-month change)

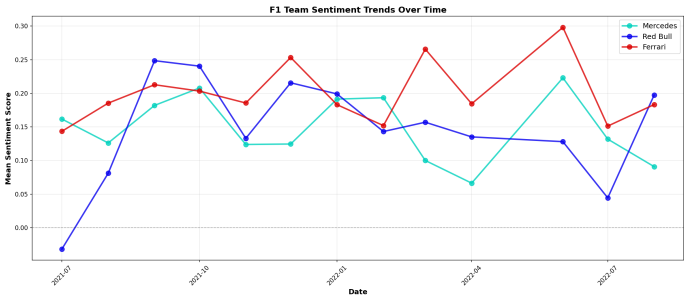

Fig. 4: Team Sentiment Trends Over Time

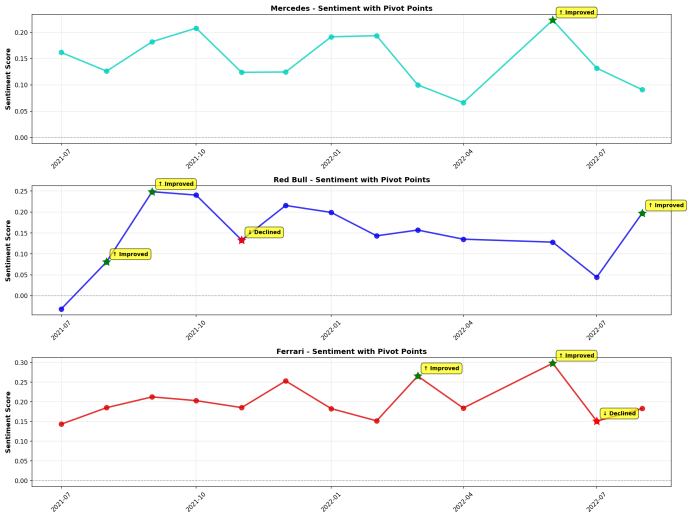

Fig. 5: Pivot Point Analysis

Sentiment trends with identified pivot points marked as red stars (↓ decline) and green stars (↑ improvement).

- Mercedes experienced a -0.12 sentiment drop in December 2021, coinciding with #AbuDhabi2021 but received a major spike in June 2022 after scoring 4 podiums in the last 5 races within that time span.
- Ferrari showed a +0.09 spike in March 2022, after winning 2 out of the first 3 races in 2022 with Charles Leclerc securing the lead of the World Driver's Championship and Ferrari leading the World Constructor's Championship. These correlations suggest NLP sentiment analysis successfully captures public reaction to major F1 events.
- Red Bull showed a +0.16 spike from August to September, following Max Verstappen securing victory in both the Netherlands and Belgium Grand Prix, thus moving him into the lead of the championship. In 2022 from July to September, Max Verstappen won 5 Grand Prix in a row, extending his championship lead over Charles Leclerc and this caused a +0.15 spike in sentiment. These correlations suggest NLP sentiment analysis successfully captures public reaction to major F1 events.

E. Summary of Key Findings

1) Preprocessing Impact::

- Standard preprocessing (removing URLs/emojis) caused 0.0333-point sentiment divergence

- Entity masking caused only 0.0007-point divergence
- 3.2% of tweets changed polarity classification with standard preprocessing
- Only 0.2% changed with entity masking

2) Team Sentiment::

- Ferrari: Most positive (+0.1937), least polarized
- Red Bull: Middle (+0.1597)
- Mercedes: Lowest (+0.1368), most polarized (26.4% negative)

V. CONCLUSION

The insights gained from this study validated the VADER model in a real-world scenario and offered a deeper understanding of emotional dynamics in the context of F1 racing. The findings contribute significantly to comprehending F1 fan behavior on social media,laying a solid foundation for developing more effective marketing and communication strategies in sports.

Future research in sentiment analysis, particularly within F1 social media contexts, presents promising directions, including repeated experiments for statistical analysis and applying Named Entity Recognition to link sentiments with specific F1 entities. Advanced techniques like PCA could be explored for deeper data insights. The development of models to identify a range of emotions and the utilization of sophisticated language models like Bidirectional Encoder Representations from Transformers (BERT) and (Generative Pre-trained Transformer) GPT offer potential for significant advancements. These efforts aim to enhance understanding and practical applications in sentiment analysis, addressing the complex emotional dynamics in the high-engagement arena of F1. In conclusion, this study provides valuable insights into the sentiment dynamics within the F1 fan community, reflecting the challenges and potential of using rules-based sentiment tools in social media sentiment analysis particularly in emotionally charged and dynamic contexts like motorsport events.

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